

Review

Advanced Algorithms in Battery Management Systems for Electric Vehicles: A Comprehensive Review

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Abstract: Electric vehicles and hybrid electric vehicles (EV) are increasingly common on roads today compared to a decade ago, driven by advancements in technology and a growing focus on sustainable transportation. These vehicles are powered by rechargeable lithium-ion batteries. A battery management system (BMS) is indispensable for ensuring the optimal performance, safety, and longevity of the EV's batteries. In this review, the latest algorithm trends for BMS software are discussed. This work also focuses on several key functionalities of BMS like the state of charge (SOC) estimation, state of health (SOH) monitoring, state of energy (SOE), and state of power (SOP). Advanced algorithms for BMS are comprehensively reviewed, including those designed for specific functionalities, as well as those developed based on existing optimization, artificial intelligence, and estimation algorithms. These algorithms address critical challenges such as maintaining symmetry during charging and discharging, preventing thermal runaway, and managing battery faults in EV systems. This work provides valuable insights for researchers and practitioners in the field of EV design and development, particularly those focusing on the advancement of BMS technologies.

Keywords: battery management system; electric vehicle; hybrid electric vehicle; lithium-ion batteries



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1. Introduction

In the past decade, electric and hybrid electric vehicles have gained popularity in the consumer market. Their reduced dependency on nonrenewable fossil fuel has demonstrated their effectiveness in promoting environmentally friendly transportation systems. However, electric vehicle (EV) technology is not a new technology. It has been around since the 19th century. The first successful electric automobile was created in 1890 with a speed of 14 miles per hour. Increasing interest in electric vehicles continued until the invention of motor vehicles that operate on gas and steam. These gas and steam powered vehicles had more speed than electric cars, which became the primary reason for the decline in interest in electric cars then [1]. Awareness of environmental impacts and the depletion of fossil fuels have driven the automotive industry toward electric and hybrid vehicles in the modern era [2]. With advancements in technology, the performance of these vehicles is now on par with conventional vehicles powered by fossil fuels.

An EV is propelled by an electric motor compared to an internal combustion engine, which derives its power from the combustion of a fuel-and-gas mixture. The most prevalent power source for EV is cells based on lithium chemistry [3]. The battery cell can be recharged

by plugging the electric vehicle into a power source. Batteries for EV are considerably distinct from those found in consumer electronics such as laptops and cell phones. They must manage high-power outputs, sometimes in hundreds of kilowatts, and store large quantities of energy, up to tens of kilowatt-hours, while fitting within the space and weight constraints of a vehicle and remaining cost-effective. The battery pack of an EV is used to store the electrical energy used to power the vehicle's motor. Extensive research and substantial investments have been devoted to the worldwide development of advanced battery technologies suitable for EVs [4].

Battery management system (BMS) measures and monitors the vehicle's speed and battery power consumption. BMS also monitors the battery's state of charge (SOC), state of health (SOH), state of energy (SOE), state of power (SOP), temperature, overcharging, and over-discharge. BMS sends a signal to the electronics control unit whenever the battery is over discharged. This unit generates control signals for the converter and regulates the motion speed. This procedure protects and lengthens the battery's lifespan while enhancing the performance and efficacy of electric vehicles [5]. BMS is very important system for EV. Therefore, this work will focus on BMS and related works.

This review is presented in five sections. The next section will present the lithium-ion battery, which is the most used battery for EV. This is followed by a detailed review on BMS. In Section 4, the algorithms for BMS are studied. The paper is then concluded in Section 5.

2. Lithium-Ion Battery

Lithium-ion batteries are commonly used because of their benefits and outstanding reliability. Lithium-ion batteries have a longer lifespan, high energy density, and a low rate of self-discharge. It takes approximately 30 to 100 kilowatt-hours (kwh) of battery capacity for an electric car to travel a certain distance [6].

A lithium-ion battery (LIB) comprises a cathode and anode divided by a porous separator saturated with electrolytes. One of the most often utilized active positive electrode materials for cathode in contemporary lithium-ion batteries is lithium ferrophosphate (LiFePO_4 /LFP). The LFP structure comprises densely packed oxygen atoms, with lithium situated in octahedral gaps, while phosphorus establishes covalent connections with oxygen, hence enhancing structural stability and battery performance. Currently, batteries utilizing lithium iron phosphate (LFP) are acknowledged as one of the most efficient commercial solutions for power [7]. This is attributable to the rapid diffusion rate of lithium ions, stable cyclability resulting from the two-phase intercalation–deintercalation mechanism within the LiFePO_4 structure, minimal alterations in specific characteristics, as well as high thermal resistance and environmental friendliness. Yet the primary drawbacks of LFP batteries are their lower electronic and ionic conductivity, as well as their lower specific characteristics [8].

Due to their high charge density and minimal weight, lithium-ion batteries have proven to be the battery of choice for EV manufacturers. Additionally, lithium-ion batteries have unique characteristics such as high energy density, low maintenance, cost, no sulfation, lack of memory effect, no requirement for periodic deliberate full discharge, the ability to accepting high charging and discharging rate, high depth of discharge, and their rate of self-discharge is less than half that of NiCd and NiMh batteries [9]. To obtain the best performance out of a battery pack, it is also necessary to entirely charge and discharge all the cells simultaneously while maintaining the same voltage [2] and, to prevent damage to the pack, the batteries must be strictly monitored for overcharge and discharge. Due to the fact that the battery charge of an electric vehicle consists of numerous cells, each one of which must be individually monitored to guarantee its safety and efficient operation,

this process becomes more complex and requires a specialized dedicated system known as the BMS.

The most typical main operational parameters recorded by BMS are voltage, current, and the interior temperature of the battery during charging and discharging. BMS is important in ensuring balance and symmetrical operation of the batteries. In the event that any of the parameters reach levels that are greater than those that have been predetermined by the safety zone, the system will transmit inputs to the protective devices, which will then cause alarms to be generated and will disconnect the battery from either the load or the charger [10].

A simulation study of a multi-chemistry battery management system to enable real-time switching for enhanced operational efficiency has been conducted in [11]. The suggested method seeks to enhance efficiency and performance by utilizing two batteries with distinct characteristics. The predominant battery chemistry employed is lithium titanate oxide (LTO) batteries, which may deliver elevated instantaneous power during periods of high-power demand. The second battery chemistry is lithium-iron-phosphate (LFP) batteries, which exhibit superior durability owing to their elevated energy density. This technique is more economical than building a system with fast-switching circuit parts. The multi-battery mechanism comprising LTO and LFP chemistries demonstrated superior performance compared to a battery pack utilizing only LFP chemistry with identical specifications. The multi-chemistry BMS offers a substantial enhancement in performance and efficiency.

Although power lithium-ion batteries are widely used, there are also numerous problems associated with their usage, such as overcharging and discharging leading to high battery temperature and explosion, inconsistent battery pack problems leading to low energy utilization, the battery cycle life being shortened, and the discharge performance being degraded under high and low temperature conditions, etc. These factors highlight the need for an efficient BMS, which has become an essential research direction in order to solve the issues associated with lithium-ion batteries and enhance system safety [12].

3. Battery Management System

A BMS is a type of control system that is specifically intended for maintaining the battery system [6] and assumes a critical role in monitoring and managing the well-being and operational efficiency of battery packs. The system collects essential information such as voltage, current, temperature, and coulomb counts in order to evaluate the real-time state of the battery. BMS anticipates the number of cells taking into account factors like temperature, voltage stability, current flow, etc., and guarantees that the batteries are operating safely. Figure 1 below illustrates the overview of BMS.

Common functionalities of a battery management system (BMS) include SOC estimate, SOH estimation, temperature measurement and estimation, cell balancing, fault detection and diagnosis, and thermal management [13]. A BMS is comprised of a variety of actuators, sensors, signal lines, and controllers. Through the examination of those data, the BMS is able to make necessary adjustments to its operations, limit the flow of energy, and notify users about any concerns. This proactive approach effectively mitigates safety risks and prevents any potential harm or damage.

The BMS was created to detect and respond to inappropriate battery behavior or readings. If the BMS detects irregularities, such as abnormal voltage or temperature levels, it can detach the battery from the load or charger or safely shut down the battery. This protection helps prevent further battery degradation and ensures the safety of the EV and its occupants. Protection of the battery is a vital aspect of BMS for EV [14]. The configuration of a battery pack for an EV depends on the desired output voltage, power,

and energy capacity. To meet these requirements, many cells are linked in series, parallel, or a combination of both. The study of battery operation and its input-output characteristics can be facilitated by the use of battery modeling. In order to get a well-designed BMS that effectively controls and optimizes battery performance, it is imperative to utilize an appropriate battery model [15]. Cell balance is yet another useful function provided by BMS. Individual cells inside a battery pack do not operate simultaneously. Some cells may be feeble, while others may be robust. The stability of the entire pack is impacted if one cell fails [5].

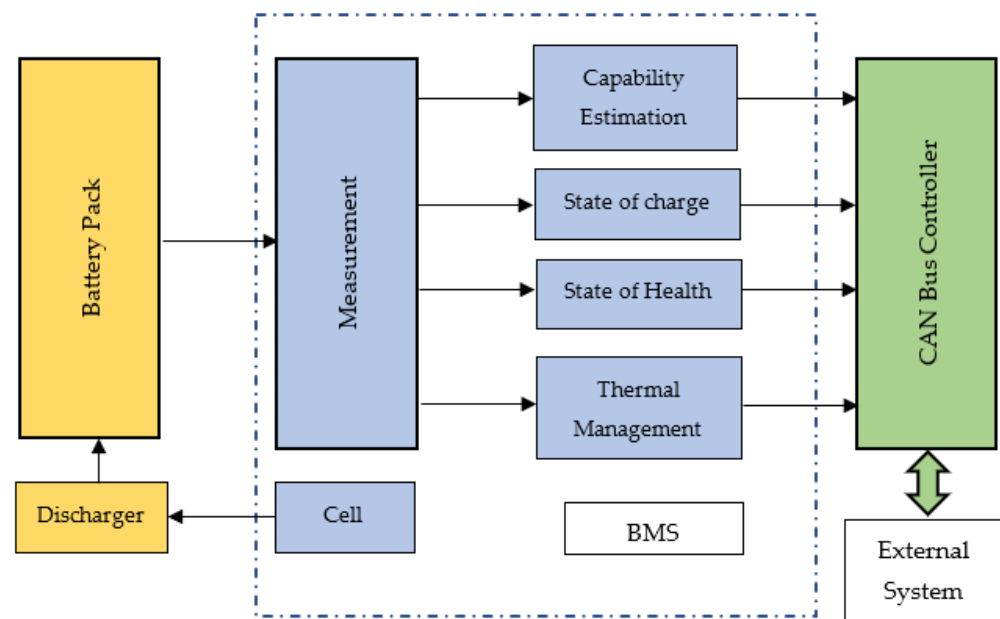


Figure 1. BMS overview diagram.

For adaptation to scale through diverse applications, ranging from small to large configuration-12 V, 100 W, and so on, for portable applications and up to 400 V, 50 kW, and even higher configuration electric vehicles-BMS, needs to be highly scalable. There are two basic types of BMS, namely centralized and decentralized, which provide a range of options for system design to accommodate various application needs [16]. The centralized type monitors a series of connected battery cells to detect issues such as overcharging, over-discharge, and thermal runaway. Decentralized type delivers management circuits to each cell or a set of cells, allowing for greater control over the battery system in terms of efficient energy management, fault handling, and special balancing procedures [17]. Modular and decentralized BMS architectures can guarantee scalability more easily: since control is distributed to individual modules, management of large-sized batteries could also become correspondingly more efficient. For instance, ref. [12] presented the potential of digital twin-based designs in scalable BMS for effective monitoring and control of large-capacity systems.

Users frequently depend on the BMS to facilitate efficient charging and discharging procedures, as well as to resolve issues such as incomplete charge or the necessity for battery rebalancing. Existing systems monitored and identified the battery's condition and alerted the driver via a battery indicator on the vehicle's display [18]. The controller circuit's primary function is to estimate the SOC, SOH, SOE, and SOP of batteries via progressive algorithms and analog signals [19]. This is illustrated in Figure 2.

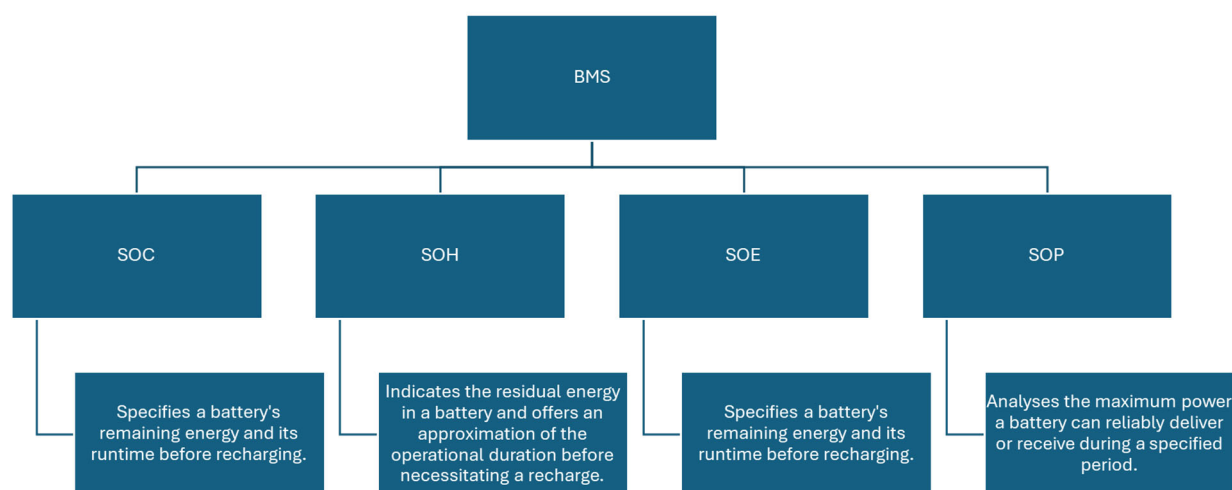


Figure 2. Battery management system.

3.1. State of Charge

The state of charge of batteries in a BMS is comparable to the fuel indicator of a conventional fuel automobile. The primary function of the SOC is to inform the driver of the battery's condition and prevent problems such as overcharge and over-discharge [20]. SOC estimation has been a significant concern for all energy-storing devices. SOC estimation provides information and precise estimates of battery dependability [21].

The battery SOC is a statistic that is commonly used to determine the amount of energy left in a battery and gives the user an estimate of how long a battery will perform before it has to be recharged. Precise battery estimation ensures the reliability and safety of electric vehicle operation. The SOC of a battery is the ratio of the battery's remaining capacity to its total capacity [22]. SOC at time t ($SOC(t)$) is typically characterized as the ratio of a battery's current capacity ($Q(t)$) to its nominal capacity (Q_n), as shown in Equation (1).

$$SOC(t) = \frac{Q(t)}{Q_n} \quad (1)$$

The manufacturer specifies the nominal capacity, which represents the maximum quantity of charge that can be stored in the battery [23].

3.2. State of Health

State of health assessment can be categorized into three classes; data-driven estimation, model-based estimation, and direct calibration-based procedures. The data-driven methodology does not necessitate extensive understanding of interior chemical characteristics and may effectively capture the intricate nonlinear interactions between SOH and measures. The accuracy of the estimation is reliant upon the quality of the data and the efficacy of feature extraction. SOH with big data facilitates the monitoring of utilized power batteries and aids in the disposal of electric car batteries post-scraping. The extensive data collecting system will retain substantial volumes of data. Filtering effective data is crucial for assessing SOH. Furthermore, the data amassed by big data platforms differ from that obtained from laboratory research. During actual vehicle operation, the operational conditions of power batteries will significantly fluctuate due to variations in driver behavior and driving surroundings [24].

The second type, called model-based estimation, is commonly utilized in BMS, which necessitates mathematical equations, empirical fits, or circuit parts to characterize the dynamics and degradation mechanisms of batteries. With an increase in battery charging and discharging times and the accumulation of time used, the battery's health status pro-

gressively deteriorates, its power and capacity demonstrate varying degrees of attenuation, the battery's capacity decreases, and its internal resistance rises. Consequently, capacity and internal resistance are typically used to define SOH [25]. SOH is an indicator of diminished performance due to degeneration in lithium-ion batteries. SOH can be calculated using Equation (2), where C_{actual} is the actual capacity of the battery and C_{fresh} is the rated capacity listed on the manufacturer's datasheet.

$$SOH(t) = \frac{C_{\text{actual}}}{C_{\text{fresh}}} \times 100, \quad (2)$$

According to [26], a battery is in a fault state when its SOH hits 80%, but ref. [27] considered 100% to 90% of the rated capacity as normal, 90% to 80% as a warning, and less than 80% as a defect.

The third category comprises direct calibration-based procedures and alternative fusion techniques. The ampere-hour (Ah) integration method necessitates entire charging and discharging cycles and is typically employed in offline contexts. The impedance spectrum-based technique does not necessitate complex calculations and fittings yet demands precise spectral excitation conditions [28]. The three categories of SOH methods are tabulated in Table 1.

Table 1. SOH estimation categories.

Method	Advantages	Disadvantages
Data-driven estimation	Needs large data training data	Adaptive
Model-based estimation	Needs accurate model	High Accuracy
Direct calibration-based procedures	Demands precise spectral excitation conditions	Necessitates complex calculations

3.3. State of Energy

SOE is defined as the ratio of the remaining energy in a battery under certain operating conditions, including fluctuating load and temperature, to the total available energy of the battery. The definition of SOE can be expressed by the following function [29].

$$SOE_t = SOE_0 - \frac{\int_0^t p(\xi) d\xi}{E_n} = SOE_0 - \frac{\int_0^t i(\xi) u(\xi) d\xi}{E_n} \quad (3)$$

where SOE_t represents the SOE value at time t , SOE_0 signifies the initial SOE value, E_n symbolizes the total available energy of the battery, and $p(\xi)$, $i(\xi)$, $u(\xi)$ indicate the power, current, and voltage at time (ξ), respectively. Numerous methodologies for SOE estimate have been developed. The power integral method was the most dominant technique among them, as it allows for the direct measurement of battery usage energy [30].

3.4. State of Power

The SOP defines the maximum power that a battery may consistently provide or absorb over a specified duration. The assessment of SOP is a crucial component of a battery management system, as it enables electric vehicles to operate stably and reliably within specified parameters. SOP indicates the battery's capacity for quick charging or discharging within its design parameters [31]. SOP estimate strategies can be categorized into two groups: characteristic mapping (CM)-based methods and equivalent circuit model (ECM)-based methods. The CM-based approaches utilize the established static interdependencies among battery power capacity, SOC, temperature, duration of the power pulse, required power, and battery voltage. The ECM-based SOP estimate techniques have gained prominence in the past decade owing to their capacity for an accurate representation of

the battery's internal dynamics, robustness, adaptability, and comparatively economy processing requirements [32].

4. Algorithms for Efficient BMS

Many advanced algorithms have been introduced. Mainly the algorithms can be divided into two categories, the first category is model-based approaches that include physics-based, simulation-based, optimization-based, estimation-based and fault diagnostic, the second categories are artificial intelligence based. Additionally, IoT-based and wireless technology had also been used for BMS.

4.1. Model-Based Approaches

Jeong et al. in their work [33] introduce a BMS algorithm that can identify and calculate battery voltage, current, and temperature. A method for SOC estimate that integrates open-circuit voltage (OCV) with the Coulomb counting method (CCM) is presented to address the limitations of both OCV and CCM. This estimate approach necessitates minimal processing resources and can enhance estimation precision. The suggested algorithm employs the OCV equation, incorporating the internal resistance and efficiency of the battery. The equation may compute the battery's charge–discharge by precisely accounting for the initial value of the CCM by the application of OCV, while taking into consideration the battery's state. Utilizing the battery efficiency formula, a predictive model for the SOH of a battery, which is contingent upon the charging duration necessary for safe operation, is integrated into the BMS algorithm to enhance reliability. During calculating a battery's efficiency, the charge–discharge current is measured to establish whether the ESS is in the charge–discharge condition. Based on the findings, the suggested method is capable of improving not only BMS performance but also battery safety by utilizing a problem diagnosis algorithm with precise SOH calculation. The study does not comprehensively compare with other parameter BMS algorithms nor evaluate its real-time efficacy.

Manas et al. [34] presented a centralized BMS for an evolving EV transportation system. Through simulation, the suggested BMS architecture and test findings are verified. The proposed BMS can control battery charge level, prevent overcharging and discharging, and maintain temperature protection, according to benchmarked testing results for the voltage, current, and temperature sensors. The benchmarked hardware and simulation results show that the suggested congregated BMS architecture can control temperature, avoid excessive charging or discharging, and maintain stability among the battery cells within specific battery parts. It lacks a comprehensive assessment of its abilities and adaptability in real-time applications.

Khalfi et al. [35] developed a model for a lithium-ion battery cell that takes consideration of the dynamic behavior of the battery cell in different settings and driving styles of electric vehicles on the highway, rural and urban roads. The model is composed of a second-order Thevenin model with six parameters. This work used the Trust-Region-Reflective and Levenberg–Marquardt algorithms in conjunction with a nonlinear least squares approach to predict the parameters of battery cells. As a measure of model performance, the mean square error (MSE) was used to validate the model; for the UDDS drive cycle, it is 3.8260×10^4 , whereas for the LA-92 drive cycle, it is 6.2674×10^4 . The Levenberg–Marquardt algorithm model is the best option because it takes fewer iterations than the other model method. This research is deficient in a thorough investigation of the model's performance during extreme or prolonged usage situations, such as high-stress driving or battery degradation.

Nivedita et al. [36] examined two distinct strategies for cooling cells, namely air cooling and direct liquid cooling. The researchers proceeded to compare the outcomes of these

procedures with the static cell temperature. In order to assess their efficacy, the effectiveness of these methods was evaluated using a conventional large-capacity lithium-ion pouch cell specifically designed for electric drive vehicles. This evaluation was conducted from the standpoint of coolant parasitic power consumption, maximum temperature increase, temperature differential within the cell, and the supplementary weight required for the cooling system. The findings indicated that an air-cooling system exhibited higher energy use in order to maintain an equivalent average temperature. The direct liquid cooling system has the most minimal increase in maximum temperature. However, it does not address cooling system reliability and maintainability, particularly coolant degradation and system damage in real-world electric vehicles.

Kumar et al. [37] used SIMULINK to develop a cost-effective BMS that constructs and simulates the circuit, ensuring optimal efficiency and security of the battery. The circuit is designed to monitor the battery's essential characteristics over a period of time, hence enhancing battery longevity and facilitating more efficient battery utilization. According to the findings, it is possible to significantly reduce greenhouse gas emissions. However, it lacks a comprehensive examination of the system's performance in real-world circumstances, specifically regarding its efficacy under diverse operating conditions.

Yan Ma et al. [38] suggested a hierarchical optimization technique for BTMS based on ant colony optimization-fuzzy sliding mode control (ACO-FSMC) to maintain the battery pack temperature at the target temperature and guarantee the cruising range of EVs. The technique for hierarchical optimization is intended to achieve control over the battery cooling rate in addition to the pump and compressor speeds. According to the simulation findings, this strategy's maximum temperature deviation and energy usage were, respectively, 43.2% and 23.0% lower than fuzzy PID. Nevertheless, the computational difficulty and feasibility of implementing the ACO-FSMC approach in real-time applications were not considered.

Hamednia et al. [39], presented a proposal for enhancing the grid-to-meter energy utilization of a battery electric vehicle through optimal battery thermal management (BTM), charging, and eco-driving techniques. An optimization problem was formulated with the objective of finding the optimal trade-off between trip time and charging cost. The problem that has been formulated was subsequently converted into a hybrid dynamical system. In this system, the dynamics in the driving and charging modes were represented using distinct functions and involve separate state and control vectors. The suggested algorithm's performance was evaluated on a hilly road, with two charging options considered along the driving path. Trip duration, including driving and charging times, decreased by 44% when compared to a situation without active battery heating or cooling. Nevertheless, it did not evaluate its scalability and adaptability over diverse real-world scenarios, encompassing varying road conditions, extreme weather, and numerous vehicle models.

Zhu et al. [40] introduced current integration method with open-circuit voltage method to calculate the average SOC. In this study, a BMS experimental platform was built that supports three serially coupled 3400 mAh lithium cobalt oxide batteries. It features precision in voltage and current measurement, SOC calculation, balancing control, and Liquid Crystal Display (LCD). The experimental results demonstrated that the cumulative error after 80 min charging by employing the combined technique is 1.1%, which may be further reduced to 0.6% with effective open-circuit voltage correction. Additionally, passive equilibrium control (PEC) was practical due to the simplicity and inexpensive cost. The results further demonstrated that PEC can serve successfully in strengthening the consistency of a battery pack over time. The discussion lacks the scalability of the methodology to larger battery packs and various operating and environmental circumstances, which are crucial for broader applicability in real-world scenarios.

Fault diagnostic algorithms are necessary for BMS. The primary function of these algorithms is to identify defects at an early stage and promptly implement suitable control measures for both the battery and the users. While operating a lithium-ion power battery, there is a risk of fire, explosion, smoke, high-voltage electric shock, and other risks. These hazards can occur due to numerous factors, including overcharge, overcurrent, and excessive temperature in the cell. These conditions can result in the formation of lithium dendrites or excessive heat generation. This can lead to the puncturing or decomposition of the solid electrolyte interphase (SEI) film, triggering a chain of negative reactions, ultimately resulting in a short circuit in the cell [41]. Xu et al. [42] developed a robust multi-objective nonlinear fault detection observer for lithium-ion batteries that is able to handle disturbances yet is very responsive to multiple faults in the battery. A comprehensive three-step multi-fault detection technique was devised to identify many types of battery defects, such as short-circuit faults, as well as faults in current and voltage sensors, using adaptive thresholding. The effectiveness of the suggested approach was further confirmed through multiple experimental case studies, which included various faults with different levels of severity and incorrect state of charge initialization. This research lacked real-time performance analysis at various driving cycle, which is crucial for dynamic BMS.

Si et al. [43] presented a work based on estimator algorithm the extended Kalman filter-amp-time integration-open-circuit voltage method (EKF-Ah-OCV), which improves the amp-time integration method and open-circuit voltage method primarily by utilizing the correction properties of the Kalman filter algorithm. The accuracy of the voltage method not only fixed the problem of the ampere-time integration method's incorrect SOC initial value estimate, but it also fixed the issue of SOC estimation mistakes that have built up over time because of inaccurate long-term current measurement. The algorithm demonstrated great performance in the complex environment as assessed by SOC and was capable of fulfilling the power demands of lithium-ion batteries. The study does not address the computationally intensive or scalable aspects, which may limit its applicability for extensive, real-time applications in extensive BMS.

In addition to these models, filters can be incorporated to eliminate model uncertainties and attain satisfactory estimation performance, such as the adaptive unscented Kalman filter (AUKF) [44] and fuzzy robust two-stage unscented Kalman filter (FRTSUKF) [45]. However, both studies lack of real-time evidence to proof their efficiency.

Guo et al. [46] estimated the state of power (SOP) using a second-order partial-adaptive fractional-order model (PA-FOM). An unscented Kalman filter (UKF)-based iterative approaching algorithm (IAA) was developed to determine the peak discharge/charge current for online SOP estimation, avoiding model linearization in the prediction windows and attaining high accuracy across the entire battery operating range. The validation results indicated that the efficacy of online SOP estimation was drastically enhanced, achieving an accuracy of superior to 0.6 W in MAE and 0.7 W in RMSE across two EV driving profiles. However, it lacked comprehensive validation of the reliability of the batteries in relation to long-term aging and degradation effects throughout prolonged operational periods.

Liu et al. [47] suggested an enhanced genetic particle filter (IGPF) to estimate the SOP based on the estimated SOC. The constraints that account for the SOC, voltage, and instruction manual were utilized to assess the discharging SOP and charging SOP. The values of discharge SOP reduced gradually from a substantial value. The charging SOP values progressively increased to a stable level. These evolving trends aligned with the current circumstances, demonstrating the accuracy of the assessments. However, this study does not examine it under extreme temperature conditions, high currents, and aged batteries for the complete verification of its robustness.

4.2. Artificial Intelligence

Artificial Intelligence techniques have been popular among the researchers of this field. Dang et al. [48] proposed a class of differential equation-informed neural networks (DENNs) for estimating battery SOC, including differential equation-informed multilayer perception (DE-MLP), differential equation-informed recurrent neural network (DE-RNN), and differential equation-informed long short-term memory (DE-LSTM). The suggested approaches not only incorporate physical principles into data-driven methods for estimating SOC but also utilize DENNs to address an inverse problem by estimating the unknown parameters of the differential equation and network parameters simultaneously. The proposed approaches provide a viable option for precisely measuring SOC in batteries, which is critical to maintaining battery dependability and longevity in a variety of applications, including electric vehicles and renewable energy systems. However, this research does not consider other factor such as thermal condition and SOH.

Zhang et al. [49] proposed temporal convolution networks for SOH predictions. The results demonstrated the accuracy and dependability of the proposed method. The proposed method outperformed traditional methods such as ampere-hour counting, which usually suffers from accumulation of errors with extended use. The research also discusses the uncertainty expression of the SOH evaluation and the use of sampling to determine the prediction results' confidence interval. However, the evaluation of SOH was based on a specific dataset (NASA public data set), which may restrict the generalizability of the results. It would be advantageous to validate the proposed method on a wider variety of datasets in order to evaluate its robustness and applicability in various scenarios.

Dineva [50] conducted an extensive study of the effectiveness of advanced machine learning techniques in predicting SOC with regression, specifically in the presence of dynamic loads. The results validate the significant benefit of advanced ML models in capturing crucial correlations among the variables of interest. The proposed method is effective in forecasting state of charge in battery systems. This is because it has the ability to store historical data and capture the cell dynamics, which is crucial for accurately predicting future charge levels. In another research [51], the ML approach was used to predict the remaining useful life (RUL) of lithium-ion batteries, which is critical for the proper management and maintenance of electric vehicle (EV) battery systems. Several models, including the regression model, the random forest regression model, the logistic regression model, the random forest classification model, and the support vector machine model, were compared to determine their accuracy. Despite the improvements, this paper has a limited parameter for validation and required excessive data for training.

Venugopal [52] suggested using an independently recurrent neural network (IndRNN) to estimate the SOH of lithium-ion batteries in EV. The goal of this study was to figure out exactly how long the batteries have left and when they will fail so that they can be used safely and reliably. The suggested method worked since the experimental results showed that it has a lower mean square error rate than similar architectures for recurrent neural networks. However, there is a dearth of discussion of potential difficulties or disadvantages of applying the suggested IndRNN technique in actual situations. Another drawback is that the suggested method was not compared to non-data-driven methods, like physics-based models, which would have given a more complete picture of how effectively it worked.

Meanwhile, Wang et al. [53] also utilized machine learning techniques, specifically long short-term memory (LSTM), to predict SOH. For this model training, 70% of the initial data were allocated as the training set, while the remaining 30% were designated as the prediction set. The neural network exhibited precise prediction outcomes as a consequence of parameter tuning. The model demonstrated great predictive accuracy across several vehicle types and different operational periods for individual vehicles. This demonstrated that the

proposed methodology is capable of accurately evaluating the condition of health of power batteries. However, the drawback is that it necessitates significant parameter adjustments, considerable computational power, and specific knowledge for optimal functionality.

Kim et al. proposed a long short-term memory (LSTM) model to predict the SOC of lithium-ion batteries in [54]. The purpose of the research is to precisely explain the complex nonlinear behavior exhibited during the charging and discharging operations of batteries to forecast the SOC. The LSTM model was trained utilizing battery data gathered under diverse temperature and load situations. However, long-term battery aging was not accounted for, as it impacts the accuracy of SOC predictions, particularly in advanced machine learning models that using historical data. Similarly, the LSTM model was adopted in forecasting the SOC of lithium-ion batteries to improve battery management systems' performance in [55].

Thomas et al. [56] introduced LSTM, and artificial recurrent neural network (RNN) architecture in machine learning, to estimate the expected lifespan of a battery. The various machine learning models were trained and tested using the datasets from NASA. Estimation of the battery life was conducted based on cell voltage, load voltage, temperature, and charge and discharge cycles. The protection provided by the BMS circuit includes overcharge, over-voltage, over-discharge, high or low temperatures, and cell imbalance at charge and discharge. The compact design circuit has validated its functionality as batteries have been charged successfully. The integration of machine learning for battery monitoring has proved very efficient and effective in predicting the life span of a battery. However, this method is suitable only for a battery pack containing a maximum of eight cells. If a battery has more than eight cells, a greater number of Atmega microcontrollers are required.

Haripriya et al. [57] proposed a BMS designed to forecast the remaining battery charge of the electric vehicle. The degradation of the lithium-ion battery in an electric vehicle was estimated via several machine learning and deep learning algorithms. The parameters, including voltage, current, and temperature, were collected from the sensors and provided as a dataset to the LSTM, Decision Tree (DT), K-Nearest Neighbors (KNN), Naïve Bayes (NB), and Support Vector Machine (SVM) algorithms. The experimental findings demonstrate that the Naïve Bayes algorithm yielded the most favorable outcomes for real-time data, as assessed by metrics including reliability, precision, recall, and F1-score. Naïve Bayes achieved an accuracy rate of 88% and was utilized to determine the remaining battery capacity, aiding in the prediction of lithium-ion battery aging. This research, however, does not address the SOH of batteries including their thermal management.

Sunori et al. [58] developed the machine learning approaches of linear regression and SVM, which are applied in MATLAB for SOC prediction. A secondary dataset comprising voltage, current, temperature, and SOC important in the development of the predictive models. 80% of this dataset has been allocated for training, while 20% has been designated for testing. The SVM model achieved an accuracy of 94.8%, whereas the Linear Regression model demonstrated a superior accuracy of 98%. Despite the efficacy of the suggested method, other critical characteristics must also be considered, as they significantly impact SOC estimation. The research should address additional critical characteristics that may significantly influence SOC estimation, including charge–discharge cycles and internal resistance.

The study in [59] examined the utilization of hybrid reinforcement learning (RL) models by integrating two advanced RL algorithms, namely deep Q-learning (DQL) and active-critic learning, to enhance the charging and discharging operations of lithium-ion batteries in electric vehicles. The hybrid models underwent extensive evaluation through simulation and experimental validation, proving their ability to produce effective battery management techniques. These solutions efficiently adapted for fluctuations in battery

state of health (SOH) and state of charge (SOC) relative inaccuracy, reduced battery voltage degradation, and complied with complex operational limitations, including charging and discharging schedules. As a concrete step toward more dependable and sustainable electric transportation networks, the findings demonstrated how RL-based hybrid models might improve BMSs in EVs. The research did not consider the integration of the suggested hybrid reinforcement learning models with the hardware and software frameworks of current battery management systems, thus presenting an issue for practical implementation.

Li et al. [60] developed three models optimized with the particle swarm optimization (PSO) algorithm: the long short-term memory (LSTM) network, convolutional neural network (CNN), and support vector regression (SVR) for precise SOH estimation. The PSO method was utilized to optimize model parameters, resulting in the development of three predictive models: PSO-LSTM, PSO-CNN, and PSO-SVR. The findings indicated that the optimized model markedly enhanced its predictive accuracy, with the RMSE and MAE decreasing by more than 0.5%. Consequently, the minimal percentage reduction in MAPE is 38%, with R^2 above 0.8, thereby indicating robust fit capabilities and validating the efficacy of the PSO approach. These experiments confirm the PSO-LSTM model as a reliable benchmark for accurately estimating the state-of-health of lithium-ion batteries, highlighting its significant applicability in practical scenarios. However, this research should consider the effects of driving conditions such as acceleration, speed, and route variation on battery health that may impact the accuracy of the SOH prediction.

Khawaja et al. [61] highlight the advancement of SOC and SOH prediction using diverse machine learning methodologies: linear regression analysis, random forest, gradient boosting, light gradient boosting (LightGBM), extreme gradient boosting (XGB), and SVM regressors. According to the findings of this research, the discharge prediction produced by the random forest estimator exhibited a substantially superior performance with minimal accuracy loss. For instance, with an optimal R^2 -score of 0.999, the random forest regressor exhibited mean absolute error, median absolute error, and RMSE values of 0.0035, 0.0013, and 0.0097, respectively. Nonetheless, it is incomparable to other comprehensive methods, such as meta-heuristic algorithms like PSO, which may similarly require significant processing power and complexity but potentially enhance the outcomes. Additionally, several drive cycles must be conducted for the models to assess their resilience under real situations. However, machine learning algorithms require large size training data with tuning to achieve highly accurate results.

In [62], a machine learning algorithm, long short-term memory neural networks, was utilized together with an estimator algorithm the unscented Kalman filter (UKF) for SOC estimations under dynamic conditions and diverse scenarios. The results showed the good potential of the suggested methodology in accurately estimating SOC in various battery applications. However, this study only reported limited parameter for validation.

Lyu et al. [63] examined battery thermal management in advanced EV and propose an advanced hybrid BTMS architecture. The battery thermal management system integrates thermoelectric cooling, forced air cooling, and liquid cooling. The liquid coolant indirectly interacts with the battery, serving as a medium for dissipating the heat produced during operation. Forced air facilitates heat dissipation from the condenser side of the thermoelectric liquid container. An electric vehicle battery system simulation has been rigorously evaluated to determine its performance. According to experimental findings, the cooling effect is promising, and the power dissipation is acceptable. Furthermore, when 40 V is provided to the heater and 12 V to the thermoelectric cooler (TEC) module, the battery surface temperature reduces roughly 43 °C (from 55 °C to 12 °C) utilizing a TEC-based water-cooling system for a single cell with copper holder. The research lacks

an examination of the long-term durability and energy efficiency of the proposed hybrid BTMS in real-world scenarios.

Harwardt et al. [64] proposed a Proximal Policy Optimization (PPO) agent that is trained using Reinforcement Learning (RL) to balance the SOC and temperature of Li-ion battery cells equipped with an active BMS. The active BMS models and battery cell models are coded in Python, together with the training environment for the agent. The efficacy of BMS based on PPO agents as determined from hyperparameter optimization resulted in a reduction in the range of balanced values by at least 28%, and in some instances, by up to 72%. However, this study does not exhaustively show the model's effectiveness for a variety of real conditions, such as variable loading profiles, extreme temperature changes, the influence of the kinds of aging in lithium-ion systems, thus limiting the possible application.

Borah et al. [65] investigated the prediction of residual energy in a battery cell during discharge throughout a broad range of current, from low to high C-rates. This research presented a novel definition of remaining discharge energy and later conducted a systematic approach that utilized machine learning for its prediction. A machine learning method was proposed to forecast the residual discharge energy at various C-rates and predetermined voltage and temperature cut-off limits. The experimental validation demonstrated that the suggested method can forecast the remaining discharge energy with a relative error of under 3% when the current fluctuates from 0 to 8 C for an NCA cell and 0 to 15 C for a lithium-iron-phosphate (LFP) cell. However, combining machine learning with physics-based modeling introduces additional computational complexity that could be challenging to handle in practical applications.

4.3. Other Smart Technologies

Another category for BMS method is smart technologies. Internet of things (IoT) has also been applied for BMS. Krishnakumar et al. [66] proposed BMS using IoT to control an electric vehicle's battery before it overheats due to continuous ON conditions. IoT-based battery management ensures online battery performance and condition monitoring. If the specified parameters fall within the acceptable range, the vehicle remains in a secure state. If the range exceeds permitted limits, the battery and the vehicle may sustain damage. The battery's performance and voltage will be consistently monitored to prevent such situations. The battery level is displayed to the user via the IoT-enabled mobile application, allowing them to determine the remaining time till depletion. The integration of GPS allows the user to identify the nearest recharging station and the distance. This solution will assist users in avoiding hazardous situations and ensuring the safe functioning of the battery in an electric car by preserving the performance and condition of the lithium-ion battery within specified threshold values. Additionally, standardization of communication protocols and interoperability standards could further accelerate the integration of scalable and adaptive BMS designs on different EV platforms. The solution to these issues will be major enablers that help in the wide-scale adoption of next-generation BMS technologies. However, this method dependent on stable internet for connectivity.

Cao et al. [67] reported a wireless BMS (wBMS) that incorporates wireless communication technologies such as Bluetooth Low Energy (BLE), Zigbee, Near-Field Communication (NFC), Wi-Fi, and cellular networks. This method provides significantly increased flexibility in battery packing, eliminates wiring, decreases battery package weight and volume, saves service and maintenance expenses, and improves reliability by lowering the risks of connection failures in comparison to wired systems. Regardless its optimistic prospects, wBMS encounters obstacles including data security, signal interference, regulatory and standardization hurdles, as well as competition from the ongoing development of wired BMS

systems, hence diminishing the apparent benefits of wBMS. Despite their potential, significant advancements must address the challenges associated with these technologies. Data security is a significant concern, and the use of robust encryption and authentication techniques is essential; blockchain appears to be a further solution for enhancing information safety against cyber threats. Signal interference constitutes an additional challenge. It can be enhanced by sophisticated algorithms for signal processing and improved shielding to guarantee reliable transmission. The absence of standardization necessitates collaboration among industry stakeholders and regulators, to develop universal standards for wireless BMS, facilitating seamless interoperability and scalability across diverse applications [67].

The highlighted research papers collectively contribute to advancing BMS for electric vehicles, offering diverse methodologies and applications. The contributions of papers in the field of BMS for electric vehicles are noteworthy. They focus, respectively, on enhancing BMS performance through accurate SOH computation, developing cost-effective BMS with circuit simulation for longevity and efficiency, proposing a centralized BMS architecture for evolving EV systems, introducing an advanced SOC estimation, suggesting a battery thermal management system (BTMS), and presenting an IoT-based BMS for real-time monitoring, contributing to enhanced safety and awareness. From the reviewed works, it can be observed that machine learning, optimization, and estimation algorithms are important in BMS. The algorithms are used to ensure that the battery is operated optimally or in prediction of the battery performance. The works reviewed above are tabulated in Table 2, highlighting the algorithms used and the main issue solved by the algorithm.

Table 2. Advanced algorithms for BMS.

Ref	Algorithm	Advantages	Limitations
Model-based			
[33]	OCV with CCM	Identify and calculate battery voltage, current, and temperature Enhance estimation precision	Lacks comparison with other BMS parameters
[34]	Simulation centralized BMS	Control battery charge level, prevent overcharging and discharging, and maintain temperature protection	Lacks detailed evaluation of robustness and real-world conditions
[35]	Trust-Region-Reflective and Levenberg–Marquardt	Estimating the parameters of the lithium-ion battery cell in different settings and driving styles	Lacks evaluation under extreme or long-term condition
[36]	Two cell-cooling methods	To study the development of a cost-effective cooling system	Lack of long-time reliability validation and maintenance requirements
[37]	SIMULINK circuit design	Optimal efficiency and security of the battery	lacks a comprehensive examination of the system's performance in real-world circumstances
[38]	ACO-FSMC (optimization algorithm)	To maintain the battery pack temperature	Does not include the computational difficulty and feasibility of implementing the ACO-FSMC approach in real-time applications
[39]	Optimal BTM	Enhancing the grid-to-meter energy	Limited driving condition
[40]	Current integration method with open-circuit voltage	To calculate the average SOC	Lacks the scalability of the methodology to larger battery packs and various operating and limited focus on battery type

Table 2. Cont.

Ref	Algorithm	Advantages	Limitations
[42]	Multi-objective nonlinear fault detection observer	To identify battery defects	Limited driving cycle
[43]	EKF-Ah-OCV (estimator algorithm)	SOC estimation	Does not address the computationally intensive
[44]	AUKF, AEKF	uncertainty and state of charge estimation of lithium-ion batteries	Lack of real-time validation
[45]	AUKF	SOC estimation	Lack of real-time validation
[46]	PA-FOM and UKF	SOP estimation	Lack of validation for long-term operation
[47]	IGPP	SOP estimation based on SOC	Lack of various extreme testing conditions
AI based			
[48]	DENNs	SOC estimation	Precisely estimating SOC in batteries Complex model
[49]	Temporal convolution networks	Predict SOH	Limited dataset for validation
[50]	LSTM, Prophet, Random Forest, Auto ARIMA	Forecasting state of charge	Limited parameter consideration
[51]	Linear regression, logistic regression, random forest regression and classification, support vector machine	To predict the remaining useful life	Require substantial amounts of labeled data
[52]	IndRNN	SOH estimation	The suggested method was not compared to non-data-driven methods
[53]	LSTM	To predict SOH	Relies on parameter tuning
[54]	LSTM	SOC prediction	Long-term battery aging not considered
[56]	LSTM RNN	to estimate the expected lifespan of a battery	Only suitable for a battery pack containing a maximum of eight cells
[57]	LSTM, Decision Tree (DT), K-Nearest Neighbors (KNN), Naïve Bayes (NB), and Support Vector Machine (SVM)	The degradation of the lithium-ion battery in an electric vehicle will be estimated via several machine learning and deep learning algorithms	Does not address the SOH of batteries including their thermal management
[58]	Linear Regression, and SVM	ML for SOC prediction and exhibited excellent performance	Limited parameter validation
[59]	hybrid reinforcement learning	deep Q-learning (DQL) and active-critic learning	Does not consider hardware and software frameworks for the proposed model
[60]	PSO-LSTM, PSO-CNN, PSO-SVR	Indicating robust fitting capabilities and affirming the efficacy of the PSO technique	Does not consider the effects of driving conditions

Table 2. *Cont.*

Ref	Algorithm	Advantages	Limitations
[61]	linear regression analysis, random forest, gradient boosting, light gradient boosting (LightGBM), extreme gradient boosting (XGB), and SVM regressors	SOC and SOH prediction	Require large size training data to achieve accuracy
[62]	LSTM UKF	Accurately estimating SOC	Limited parameter for validation
[63]	Advanced hybrid BTMS architecture	To study thermal management of batteries	Lack of analysis for long-term durability
[64]	Proximal Policy Optimization (PPO)	To balance the SOC and temperature	Limited validation
[65]	NDCTNet	Prediction of residual energy of battery cell during discharge throughout a broad range of current	The computational complexity might cause difficulty with handling
Other smart technologies			
[66]	IoT: ThingSpeak	To control an electric vehicle's battery before it overheats	Dependency on network connectivity
[67]	Bluetooth Low Energy (BLE), Zigbee, Near-Field Communication (NFC), Wi-Fi, and cellular networks	decreases battery package weight and volume, saves service and maintenance expenses, and improves reliability	Encounters obstacles including data security, signal interference, regulatory and standardization hurdles

4.4. Challenges and Limitations

Conventional BMS methodologies have primarily concentrated on static algorithms that monitor and control battery parameters according to established models. These techniques, however, are insufficient considering the nonlinear behavior of batteries under fluctuating load circumstances and environmental factors, resulting in poor reliability and diminished battery longevity [68]. Despite technological developments, the integration of multiple computational approaches into a unified BMS framework continues to pose a significant difficulty and constraining potential advantages such as prolonged battery life, enhanced safety, and augmented energy economy.

According to [69], there are three main areas where lithium-ion battery management presents technical issues and challenges. First is the complex and nonlinear electro-thermal behaviour of lithium-ion batteries, thus complicating the modelling process. Second is the inability of sensors to directly assess the internal states of lithium-ion batteries and their high susceptibility to noise and ambient temperature. This makes it challenging to obtain a precise battery estimate. The last issue is the complicated structure when groups of lithium-ion batteries are used, which makes it hard to control precisely.

Moreover, because the work in [69] highlighted the inability to capture all the complexities in real life, inaccuracies might arise in performance predictions. Furthermore, hardware configurations and the settings of parameters can reduce their general applicability to only a few environments.

Real-time validation is one of the major challenges for BMS development. It enables effective and dependable BMS by evaluating algorithms under dynamic, real-world situations. The system's performance is to be evaluated for coverage across many scenarios, encompassing driving behaviors and environmental conditions, hence enhancing robust-

ness and precision. A decentralized yet synchronized real-world system for intelligent battery management was developed in [70] with a universal controller with cloud computing functionality, four charge regulators, and a collection of sensor-equipped battery monitors featuring networking and Bluetooth capabilities. The results demonstrated the effective performance of the decentralized BMS system in maintaining threshold values indicated in the SOC graphs under appropriate conditions, hence preventing both overload and a decline below 80%. Regardless, substantial obstacles exist in obtaining accurate measurements, especially with state of health (SOH) assessments. These limit the proper improvement of precise predictive diagnostics for accurate remaining useful life (RUL) estimation of batteries.

Real-time analysis technologies for BMS like long range (LoRa) have made significant progress [71], but there are still problems with creating accurate real-time estimations. As a result, the majority of these solutions fail to account for all the complexities that arise in real life, which leaves room for model prediction errors. Their effectiveness are limited and their dependability for real-time application greatly impacted by differences in hardware configurations and parameter settings across various situations. Resolving the issues is essential to maximizing LoRa's potential for enabling long-range, low-cost, and low-power remote battery monitoring.

Reliability and scalability can be improved by applying effective control strategies utilized in shipboard microgrids for battery management systems. Secondary distributed control methods, demonstrated in isolated shipboard microgrids, are crucial for addressing issues related to voltage and frequency fluctuations, accurate power distribution, and maintaining system stability in dynamic environments [72].

5. Conclusions

Integration of dynamic battery behavior modeling with BMS is a complicated task that requires many computer resources and may involve making changes in real time, which could make it hard to implement. Furthermore, the effort to develop robust and adaptive optimization algorithms for a wide range of electric vehicle types and driving conditions is not straightforward. This work necessitates the consideration of varied user behaviors, traffic patterns, and environmental considerations. Furthermore, it is imperative to broaden the scope of safety mechanisms and reliability assessments beyond laboratory-based tests in order to incorporate real-world situations. This necessitates the implementation of comprehensive testing and validation procedures. The examination of the environmental consequences, grid interactions, and user behavior requires the involvement of multiple disciplines and the careful consideration of regulatory and infrastructure constraints.

Despite the advancements of BMS, critical research gaps are evident. While the studies address monitoring, basic control, and specific functionalities, there is a clear need for in-depth exploration of advanced control strategies. The papers lack extensive discussion on integrating machine learning algorithms within BMS frameworks, leaving untapped potential for predictive analytics, fault detection, and adaptive control. Notably absent are discussions on cybersecurity considerations, scalability challenges, and real-world implementation issues. The lack of attention to interoperability standards and energy-efficient cooling strategies further underscores areas requiring research. Addressing these gaps is crucial for developing more robust, adaptive, and widely applicable BMS, ensuring the continued advancement and widespread adoption of electric vehicles.

BMSs, identical to other components of EVs, are continuously enhanced to align with the latest developments in the industry. Engineers globally are actively developing various components and functions of BMS. Three primary research directions can be identified:

enhancing electric vehicle adoption, integrating with renewable energy resources, and addressing cybersecurity.

Future study might explore more algorithms and compare them, including those not reported in this paper, to tackle the problems of EV BMS. The development of advanced algorithms can enhance real-time state estimation, thermal management, and energy optimization, hence improving the reliability, efficiency, and performance of electric vehicle batteries. Testing of these algorithms will focus on real driving settings, including diverse driving patterns such as urban, highway, and mixed-mode scenarios, as well as environmental restrictions to ensure robustness and practical applicability. Interdisciplinary collaborations, encompassing connections between material science, control systems, and software engineering, may yield innovative techniques.

Other important issues are such practical concerns as cost reduction, scalability, and environmental adaptability. They require research in low-cost sensor technologies, energy-efficient communication protocols, and lightweight algorithms. These would guarantee the applicability of such systems throughout diverse applications, from small-scale residential systems to large-scale industrial setups. Environmental adaptability will be assured by implementing designs of systems ensuring reliable operation in hostile conditions like extreme temperature or high humidity thus they are globally applicable. It will help support real-time monitoring and control by the convergence of advanced sensing and communication technologies, such as 5G and long-range communication systems like LoRa. The embracement of such research areas and focused attention on them will serve to set the stage for next-generation energy management systems, the efficiency and sustainability of which are fully assured.

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